Multi-view optical tomography using L1 data fidelity and sparsity constraint

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ABSTRACT

This paper describes a novel reconstruction algorithm for microscopy axial tomography, which reconstructs a 3-D volume using multiple tilted views through an off-centered aperture and numerical processing. The main contribution of this paper is a derivation of novel optimization criterion and algorithm for a cost function with L_1 fidelity term and sparsity constraint. A parallel coordinate descent (PCD) algorithm has been derived as an efficient optimization methods, which corresponds to iterative application of projection and *nonlinear* back-projection using median. Numerical simulation results using synthetic and real microscopy data show that accurate reconstruction can be obtained rapidly, and interference artifacts from high contrast objects in a volume can be removed efficiently. Our algorithm is quite general, and can be used for many other tomosynthesis applications with limited number of views.

Keywords: PCD, OSBP, tilted tomography, off-centered aperture

1. INTRODUCTION

Optical sectioning is often used for real time monitoring of three dimensional biological samples, or for non-destructive defect detection in semiconductor industry.[1] Confocal microscopy is the most popular optical sectioning technique due to its rejection of out-of-focal plane interference using matched pinholes; however, the confocal microscopy requires a time-consuming lateral and axial scanning, and exhibits low light efficiency. Furthermore, most of the confocal microscopy systems are too expensive to be used for routine monitoring tasks.

As an alternative, computational optical sectioning technique using wide-field optical microscope and computer processing is often employed. Two types of computationally sectioning methods have been investigated; the deconvolution microscopy,[1] and the microscopy axial tomography approaches.[2–6] The deconvolution approach acquires 3-D structure from multiple images with different focal lengths, whereas the microscopy axial tomography uses the series of tilted views of the samples through a rotating off-centered aperture (see Figure 1). The roles of the off-centered aperture are to reduce the numerical aperture (NA) for a larger depth-of-focus (DOF) to remove axial scanning, and to generate a series of tilted views containing the 3-D information about samples.

The imaging concept of microscope axial tomography is closely related to other imaging techniques called *tomosynthesis*, or laminography[11] in medical and industrial applications. In the digital breast tomosynthesis system, about 15 tilted x-ray projection images of the compressed breast are usually acquired, and the volumetric reconstructions are obtained using numerical algorithms to increase the sensitivity and specificity of breast cancer diagnosis.

The main focus of this paper is a novel tomosynthesis algorithm that can be used for any type of aforementioned tomosynthesis system, even though the main application in this paper is for microscope axial tomography. Our main contribution is a novel tomosynthesis algorithm, which is obtained by minimizing the L_1 data fidelity term under sparsity constraint. The basic idea of these approaches is that minimizing the L_1 data fidelity terms gives less importance to outliers or *wild* signal samples, hence can eliminate impulse noises.

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Based on aforementioned reasoning, we conjecture that the L_1 fidelity approach might be efficient in removing interference artifacts from high contrast objects in a volume, which has been considered a major artifact in limited tomography. A parallel coordinate descent (PCD) algorithm has been derived to minimize the cost function with L_1 data fidelity and sparsity constraint, which results in iterative applications of projection and *nonlinear* backprojection using order statistics. This observation gives a new insight of the order statics-based backprojection (OSBP) algorithm of General Electric (GE) for digital breast tomosynthesis, which eliminates the strong outlier before backprojection.[10] Our analysis show that the OSBP corresponds to a special case of our algorithm derived based on L_1 data fidelity term and sparsity condition.

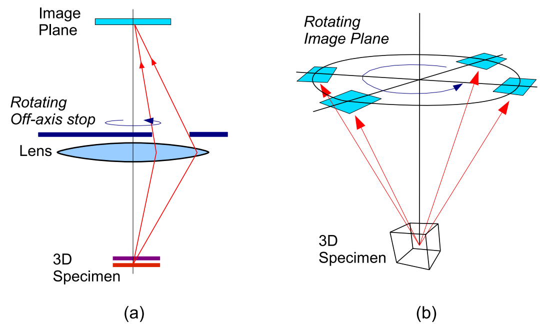


Figure 1. A microscope axial tomography[2]: (a) side view and (b) its projected view in image plane.

2. THEORY

Consider a general inverse problem, where an original scene $\mathbf{f}$ has to be recovered given its observation $\mathbf{y}$:

$$\mathbf{y} = \mathbf{H}\mathbf{f} + \mathbf{w} \tag{1}$$

where $\mathbf{H}$ is a blurring matrix and $\mathbf{w}$ denotes the additive noise. Assuming that the original scene $\mathbf{f}$ is sparse, the L_1 data-fidelity based minimization problem is given by

$$\min_{\mathbf{f}} ||\mathbf{y} - \mathbf{H}\mathbf{f}||_1 + \lambda||\mathbf{f}||_1 \tag{2}$$

where $||\cdot||_1$ denotes the L_1 norm, and λ denotes the regularization parameter. The reasoning of the L_1 regularization term to impose sparsity constraint comes originally from Lasso by R. Tibshirani.[12] Inverse problems under sparsity constraint has become active area of research in signal processing under the name of *compressed sensing* or *compressive sampling*.[13]

Recall that the conventional sparse reconstruction approach for Eq. 1 is based on L_2 data fidelity term[16]:

$$\min_{\mathbf{f}} ||\mathbf{y} - \mathbf{H}\mathbf{f}||_2^2 + \lambda||\mathbf{f}||_1 \tag{3}$$

where $||\cdot||_2$ denotes the L_2 norm. Even though the difference between Eq. (2) and Eq. (3) is only the data fidelity term, this difference results in delicate changes of the solutions.

The original approaches to address Eq. (2) is based on linear programming, which is usually computationally expensive.[17] Instead, this paper employs a parallel coordinate descent (PCD) approach, which was originally suggested by M. Elad.[14] More specifically, the PCD is a simultaneous implementation of the iterative coordinate descent (ICD) algorithm[15] that minimizes each coordinate values alternatively until the algorithm converges.[9]

Specifically, ICD algorithm updates $\mathbf{f}$ voxel by voxel by perturbing value of $\mathbf{f}$ at the i-th voxel point by the amount Δf_i, i.e. $\hat{f}_i = f_i + \Delta f_i$. Such image update results in the reduction of residual:

$$\mathbf{e} \leftarrow \mathbf{e} - \mathbf{H}_{*i}\Delta f_i, \tag{4}$$

where $\mathbf{e} = \mathbf{y} - \mathbf{H}\mathbf{f}$ and $\mathbf{H}_{*i}$ denotes the i-th column of the blur matrix $\mathbf{H}$. Then, the perturbed L_1 data fidelity term is given by:

$$\begin{aligned} ||\mathbf{y} - \mathbf{H}\hat{\mathbf{f}}||_1 &= ||\mathbf{e} - \mathbf{H}_{*i}\Delta f_i||_1 \\ &= \sum_{n\in\mathcal{X}} |e_n - H_{n,i}\Delta f_i| \end{aligned} \tag{5}$$

where $H_{n,i}$ denotes the (n, i) element of the matrix $\mathbf{H}$, e_n is the n-th element of $\mathbf{e}$, and $\mathcal{X}$ denotes the index set such that $\{n|H_{n,i} \neq 0\}$, respectively. Then, the resultant 1-D cost function with respect to Δf_1 is given by

$$C(\Delta f_i) = \sum_{n\in\mathcal{X}} |e_n - H_{n,i}\Delta f_i| + \lambda|f_i + \Delta f_i| + const. \tag{6}$$

Theorem 2.1 characterizes the minimizer f_i^* of Eq. (6).

THEOREM 2.1. *The minimizer f_i^* of Eq. (6) should come from the following discrete set:*

$$\mathcal{S} = \{-f_i\} \bigcup \left\{ \frac{e_n}{H_{n,i}} \right\}_{n\in\mathcal{X}} . \tag{7}$$

Furthermore, if $H_{n,i}$ is constant for all n and $\lambda = 0$, then the optimal solution is given by

$$\Delta f_i^* = median\{e_n\}_{n\in\mathcal{X}} . \tag{8}$$

Now the parallel coordinate descent update search direction is obtained by collecting the pointwise search direction from ICD:

$$\mathbf{d} = \begin{pmatrix} \Delta f_1^* \\ \Delta f_2^* \\ \vdots \\ \Delta f_N^* \end{pmatrix} \tag{9}$$

and the new estimate is updated by

$$\hat{\mathbf{f}} \leftarrow \mathbf{f} + \alpha\mathbf{d} . \tag{10}$$

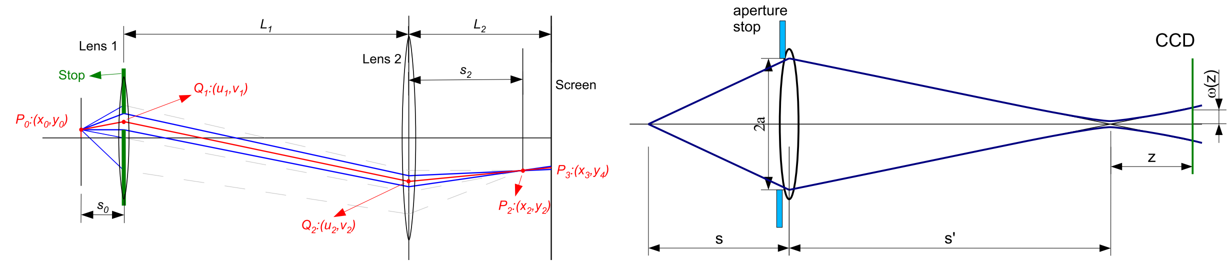
Note that the search direction Eq. (9) is not in general descent direction since we have stacked the ICD search direction into a single vector. However, if we obtain α using a line search to guarantee the decrease of cost function Eq. (2), the resultant update is along the descent direction.

The optimal step size α can be obtained easily by minimizing the following one dimensional cost function:

$$\begin{aligned} C(\alpha) &= ||\mathbf{y} - \mathbf{H}\hat{\mathbf{f}}||_1 + \lambda||\hat{\mathbf{f}}||_1 \\ &= ||\mathbf{e} - \alpha\mathbf{d}||_1 + \lambda||\mathbf{f} + \alpha\mathbf{d}||_1 \\ &= \sum_{i=1}^{N}\sum_{n\in\mathcal{X}} |e_n - \alpha H_{n,i}\Delta f_i| + \lambda\sum_{i=1}^{N} |f_i + \alpha\Delta f_i| \end{aligned} \tag{11}$$

Using Theorem 2.1, we know that the optimal step size α belong to the following finite set:

$$\mathcal{A} = \left\{ -\frac{f_i}{\Delta f_i} \right\}_{i=1,\cdots,N} \bigcup \left\{ \frac{e_n}{H_{n,i}\Delta f_i} \right\}_{n\in\mathcal{X}, i=1,\cdots,N} . \tag{12}$$



(a) (b)

Figure 2. Optical microscope tomography with off-centered aperture.

3. APPLICATION TO MICROSCOPE AXIAL TOMOGRAPHY

3.1. Geometry Optics

Let us consider the microscope tomography system in Fig. 1 with a rotational off-centered aperture. Here, a ray from the sample passes through the objective lens and the aperture before it arrives at the image plane. As shown in Fig. 1(b), the rotating off-centered aperture generates a series of tilted view images on the image plane, hence by numerically processing the tilted view images, we may obtain the 3-D structure of the sample.[2]

Fig. 2 illustrates the simplified schematic diagram of the proposed 3-D microscopy tomography with an off-centered aperture. As explained in Appendix, the geometric optic approximation of the image center location on the screen is given by

$$(x_c, y_c) = -\frac{f_2}{s_0}(u, v) - \frac{f_2\delta}{s_0}(\zeta, \eta) \tag{13}$$

where (x_c, y_c) denotes the image location on the CCD plane when the point source is located at (u, v, w), and $\delta = (|w| - f_1)/f_1$ is the proximity ratio of the samples with respect to the focal length, and (ζ, η) denotes the center coordinate of the off-centered aperture. Here, the first term corresponds to the magnification factor of a general optical system, whereas the second term is the aperture dependent term.

3.2. Point Spread Function

In general optical imaging systems, the image of a point source is not a point but a distribution due to the diffraction. The intensity distribution is called the point spread function (PSF). According to the diffraction theory,[18] the distribution of the light intensity on the image plane is given by

$$PSF(x - x_p, y - y_p, z|z_p) = \left| A \int_0^1 e^{iW(\rho,z|z_p)} J_0(krNA\rho)\rho d\rho \right|^2, \tag{14}$$

where $k = \frac{2\pi}{\lambda}$, and $W(\rho, z|z_p) = k \cdot OPD$, and $r = \sqrt{(x - x_p)^2 + (y - y_p)^2}$, and λ is the wavelength, $J_0(x)$ denotes the Bessel function of first kind. Optical path difference is imposed by the difference between ideal and actual optical systems. The physical size of one pixel is determined by pitch size in CCD camera and total magnification power.

4. EXPERIMENTAL RESULTS

We have compared our reconstruction results with those of the conventional algorithms. The specification of proposed off-centered aperture is determined by numerical aperture (NA) and depth-of-focus (DOF). The numerical aperture (NA) is relevant to focal length (f) of objective lens and aperture radius (a) by geometry optics, that is $NA = n_i \cdot (a/f)$. Depth-of-focus (DOF) is a dependent term of NA. In the proposed methods, the analytic PSF model in (14) are incorporated during the projection and back-projection to account for the diffraction phenomenon.

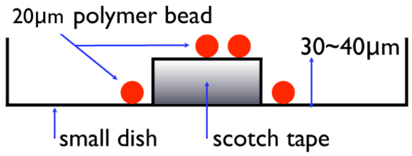
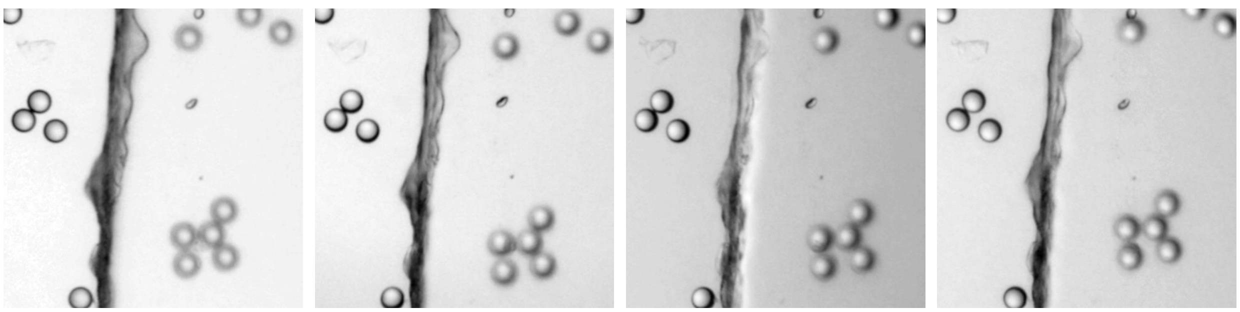


Figure 3. Experiment setup.

The proposed off-centered aperture is mounted on Olympus® IX51 inverted microscope system. Among the commercial objective lenses, a $20X$ objective lens is moderate to monitor $20\mu m$ polymer bead. In order to meet the specification of DOF of $30\mu m$, off-centered aperture radius is set to $1.5mm$.

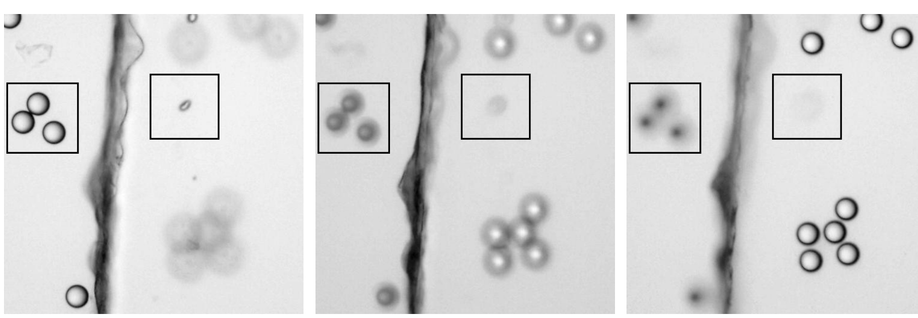
A phantom as shown in Fig. 3 is used for our experiments. The $20\mu m$ beads are located on top of the transparent tapes with thickness of about $30\mu m$ or at the bottom of the sample dish. The main motivation for this double layer structure is to verify that our algorithm can successfully resolve axially varying structures. Four tilted views are acquired at every 90^o . Since the brightness of the projection images varies depending on the view angle due to the different illumination condition, we have chosen center region of each view that has uniform brightness as shown in Fig. 4



(a) (b) (c) (d)

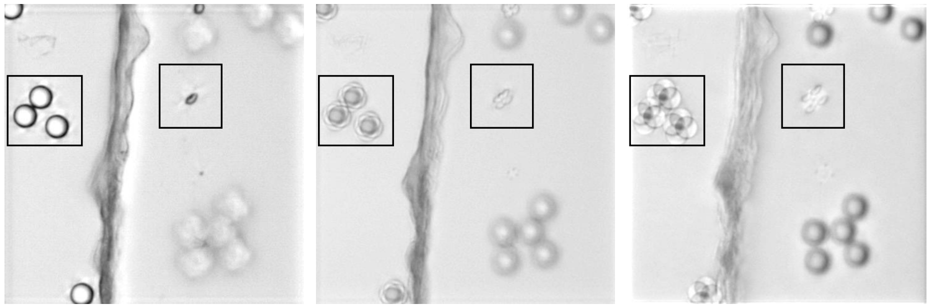
Figure 4. Four views acquired at the tilting angle of 90^o, 180^o, 270^o, 360^o, respectively.

Figs. 5(a)-(c) illustrates the images taken by IX51 microscopy at different focal lengths. Since the images are taken by the wide-field microscopy, we can still observe the out-of-focus interference. Figs. 6(a)-(c) shows the reconstructed z-slices using the conventional L_2 fidelity term at $0, 15\mu m$ and $30\mu m$, respectively. Even though each layer images are resolved, there still remain significant ghost artifacts. However, our L_1 fidelity based reconstruction results in Figs. 6(a)-(c) clearly demonstrate that the most of the ghost artifact are suppressed thanks to the rejection property of the L_1 norm.



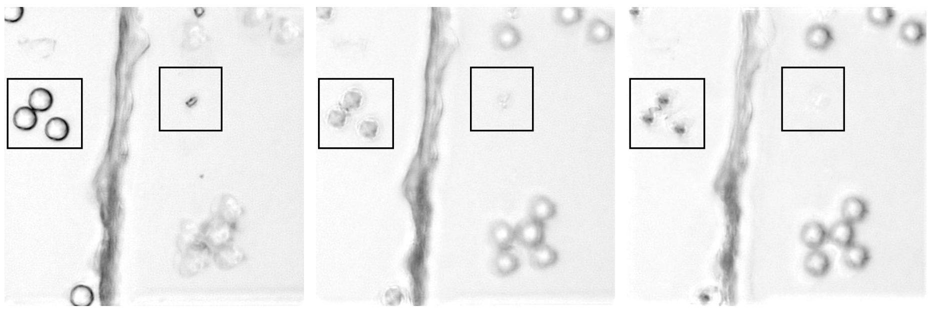
(a) $0\mu m$ (b) $15\mu m$ (c) $30\mu m$

Figure 5. IX51 microscope images at different focal lengths.



(a) $0\mu m$ (b) $15\mu m$ (c) $30\mu m$

Figure 6. Reconstructed slices using the conventional backprojection algorithm.



(a) $0\mu m$ (b) $15\mu m$ (c) $30\mu m$

Figure 7. Reconstructed slices using the proposed method.

5. CONCLUSION

This paper described a novel reconstruction method for microscopy axial tomography using L_1 data fidelity and the parallel coordinate descent (PCD) algorithm. The advantages of the L_1 data fidelity minimization is that it can effectively remove the outlier data that may cause the out-of-focus ghost artifact in the conventional back-projection algorithm. We modified IX51 microscope to have a rotating off-centered aperture to acquire tilted view images. The aperture diameter and the size was determined based on the geometric optics and diffraction theory to satisfy the specification of the depth-of-focus. Experimental results demonstrated that the ghost artifact can be efficiently suppressed using the proposed algorithm even from very limited number of views.

Appendix

Fig. 2(a) illustrates the simplified schematic diagram of the proposed 3-D microscopy tomography with an off-axis stop. The sample at $P_0 = (u, v, w)$ has its image at P_2. Our interest is to calculate the position on the screen (P_3) when a ray from the sample passes through a specific point of Lens 1. As shown in the Fig. 2(a), the ray from P_0 passes through Q_1, Q_2, and then P_2 sequentially before it arrives at P_3. Let s_1 denote the distance from Lens 1 to its image plane, and (u_1, v_1) is the coordinate in the image plane (which are not shown in Fig. 2(a)). Using Gaussian optics approximation, we have

$$\frac{1}{s_1} = \frac{1}{f_1} - \frac{1}{s_0} \tag{15}$$

$$(u_1, v_1) = -\frac{s_1}{s_0}(u, v) \tag{16}$$

where s_0 denotes the sample distance and f_1 is the focal length of the Lens 1. Here, the notation $(a, b) = \kappa(d, e)$ denotes $a = \kappa d, b = \kappa e$.

Now, we can obtain the coordinate on P_3:

$$(x, y) = \frac{L_2}{s_2}(x_x, y_2) + \left(1 - \frac{L_2}{s_2}\right)(\zeta', \eta') \tag{17}$$

and

$$(\zeta', \eta') = \frac{L_1}{s_1}(x_1, y_1) + \left(1 - \frac{L_1}{s_1}\right)(\zeta, \eta) \tag{18}$$

where L_1 denotes the distance between the Lens 1 and Lens 2. Hence, the coordinate on P_3 with respect to the sample point (u, v) and stop coordinates (ζ, η) can be represented by:

$$(x, y) = \alpha(u, v) + \beta(\zeta, \eta) \tag{19}$$

where α and β are given by

$$\alpha = -\frac{L_2}{s_0}\left[1 + \left(1 - \frac{L_2}{f_2}\right)\frac{L_1}{L_2}\right] \tag{20}$$

$$\beta = \left[\left(1 - \frac{L_2}{f_2}\right)\left(1 + \frac{L_1}{s_0}\frac{s_0 - f_1}{f_1}\right) - \frac{L_2}{s_0}\frac{s_0 - f_1}{f_1}\right] \tag{21}$$

Note that Eqs. (20) and (21) have the common term $(1 - L_2/f_2)$ which vanishes when the CCD screen location L_2 coincides the focal length of Lens 2. Under this condition, Eq. (19) becomes

$$\begin{aligned}(x, y) &= -\frac{f_2}{s_0}(u, v) - \frac{f_2}{s_0}\frac{s_0 - f_1}{f_1}(\zeta, \eta) \\ &= -\frac{f_2}{s_0}(u, v) - \frac{f_2}{s_0}\delta(\zeta, \eta)\end{aligned} \tag{22}$$

where $\delta = (s_0 - f_1)/f_1$ is the proximity ratio of the samples with respect to the focal length and we use the first order approximation in Eq. (13). Here, the first term corresponds to the magnification factor of a general optical system, whereas the second term is the aperture stop dependent term. Therefore, even if rays from sample pass through the identical coordinate on the stop, their images on the screen may be different if the distances between the samples and the objective lens are distinct.

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